2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)
16 – 18 April 2026, Chittagong, Bangladesh

2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN) | 979-8-3315-4990-9/26/$31.00 ©2026 IEEE | DOI: 10.1109/QPAIN69676.2026.11546263

Design and Performance Comparison of 4-bit Ripple-Carry and Carry-Look-Ahead Adder in 90-nm CMOS VLSI Technology

Ahasan Ullah Khalid[1], Anika Tabassum[2], Arif Istiaque Rupom[3],
Md. Farhad Hossain[4], Md. Azad Hossain[5]
[1,2,3,4,5,6]Department of Electronics and Telecommunication Engineering, Chittagong University of Engineering and Technology
Chattogram, Bangladesh
[1,2]{aukhalid02, tabassumanika0029}@gmail.com, [3,4,5]{arif.ete, farhad.hossain, azad}@cuet.ac.bd

***Abstract*—This paper presents the schematic and transistor-level design, as well as a comparative performance analysis, of two 4-bit binary adder architectures: the Ripple-Carry Adder (RCA) and the Carry-Look-Ahead Adder (CLA). Both adders were designed at the schematic, symbol, and transistor-level layout and verified through DRC/LVS using the GPDK90, 90nm CMOS Process Design Kit. A thorough performance comparison was conducted based on key parameters, including best-case and worst-case propagation delay, average power consumption, silicon area, and transistor count, with advanced parameters like the Power-Delay Product (PDP) and Area-Delay Product (ADP). The simulation results mark significant trade-offs between the two architectures. The CLA has superior speed, with a worst-case propagation delay of 79.7 ps, that significantly outperforms by approximately 3.2 times than the RCA's delay of 255.9 ps. But it also performs significantly worse in terms of transistor numbers, silicon area, and average power consumption, demonstrating the design trade-offs between the two adders.**

***Index Terms*—Ripple Carry Adder, RCA, Carry Look-Ahead Adder, CLA, VLSI, CMOS, GPDK90, 90 nm**

I. Introduction

Digital Adders are a fundamental building block in modern devices. They are really an important part as they serve as the core components in microprocessors or chips, various ALUs, and DSPs, among others [1]. Among all the adders, 2 of the most common adders are the RCA, Ripple-Carry Adder, and the CLA, Carry-Look-Ahead Adder. The RCA has a simple serial design where the carry out from each full adder stage becomes the carry in for the next full adder, creating a ripple type effect in the circuit [2]. But the problem is, this limits the performance of the circuit by the propagation delay, the time it takes to go from Cin to Cout, which increases linearly with the number of bits the adder will add [3]. On the other hand, the CLA has a more complex approach, a parallel way by generating and propagating the carry signals simultaneously in the circuit, which significantly reduces the overall delay of the whole operation [4], [5]. However, it increases circuit complexity, average power consumption, and silicon area of the adder [6], [7].

Both types of adders have been thoroughly studied before because of their central role in Arithmetic Logic Units or ALUs and Digital Signal Processors or DSPs. The Ripple Carry Adder (RCA), which is known for its structural simplicity, has long been considered the baseline architecture for 4-bit addition [8]. However, its linear carry propagation delay has been consistently reported as a huge limiting factor in high-speed systems [9]. To overcome this limitation, Carry Look-Ahead Adders (CLAs) were proposed, where generate and propagate logic were implemented to compute carries in parallel [10]. While these studies provide us with valuable architectural insights into these adders, most of them are performed at gate-level or RTL abstraction, without any detailed transistor-level layout with DRC/LVS verification and RCX extraction [11]. Some works report simulation results using behavioral models, but do not capture the practical layout design constraints imposed by modern CMOS processes [12].

This paper presents a comprehensive schematic to transistor-level design and performance comparison of 4-bit RCA and CLA architectures implemented using the GPDK90 CMOS Process Design Kit (90-nm technology). The main contributions or novelty of this paper are as follows:

1) Transistor-level layout implementation is done of RCA, and CLA with GPDK90 technology, including full DRC and LVS verification and RCX extraction.
2) Comprehensive performance analysis is performed on the basis of best and worst-case propagation delay, average power, silicon area, and transistor count.
3) The power delay product (PDP) and the area delay product (ADP) of the two adders were evaluated, providing a clearer picture and deeper insight into energy, power, and area efficiency.

The organization of this paper is as follows:

In Section II, the Design Methodology and Implementation have been explained in detail. Here, it is explained how the adders were designed from gate-level schematic to transistor-level layout. Also, the DRC/LVS verification has been done. In the next section, the Simulation Methodology has been explained. And after that, the results are analyzed, and then a discussion and a conclusion are added to summarize the study.

979-8-3315-4990-9/26/$31.00 ©2026 IEEE

Authorized licensed use limited to: INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY. Downloaded on October 25,2026 at 13:34:52 UTC from IEEE Xplore. Restrictions apply.

II. Design Methodology and Implementation

The design and implementation of the 4-bit Ripple-Carry Adder (RCA) and Carry-Look-Ahead Adder (CLA) were implemented using the GPDK90 CMOS Process Design Kit (PDK). The methodology was structured into 3 main stages. For each Adder, these 3 main stages have been completed to get the final result. These 3 stages are:

1) Gate-level Schematic design: For each adder, first, a Gate-level schematic has been designed.
2) Transistor-level Full Layout design: From the schematic, a detailed transistor-level full layout has been designed.
3) DRC/LVS verification and Simulation: The full transistor-level full layout has been verified using the DRC/LVS verification, and then it was simulated using RCX extraction.

All the stages are explained with proper diagrams for each adder.

A. 4-Bit Ripple-Carry Adder (RCA):

The Ripple Carry Adder (RCA) was designed using 4 Full Adders, each implemented with 9 NAND gates. At first, the gate-level design for NAND gates was implemented. The logic expressions for the full adder are given as:

$$\text{Sum} = A \oplus B \oplus C_{in} \tag{1}$$

$$C_{out} = (A \cdot B) + (C_{in} \cdot (A \oplus B)) \tag{2}$$

1) Full Adder Design and Implementation: The Full Adder was made using the NAND Gates. First, the Schematic and then the transistor-level layout of the Full Adder were made, verified through design rule check (DRC) and layout versus schematic (LVS), and simulated in ADE L using RCX extraction.

a) Full Adder Schematic: Here, the full adder schematic was made using 9 NAND gates (Fig. 1). The inputs are A and B, and the outputs are Y and Cout.

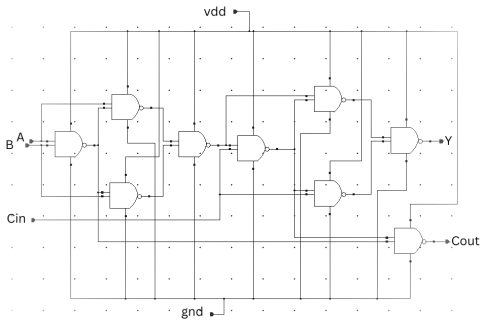


Fig. 1: Schematic of the Full Adder

b) Full Adder Layout: Here, the full adder transistor-level layout was made using standard cell NAND gate cells (Fig. 2). The DRC and LVS check was done, and finally, it was ready for RCX extraction and simulation.

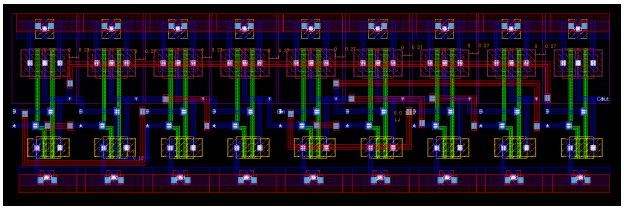


Fig. 2: Layout of the Full Adder

2) RCA Design and Implementation: The RCA was made using the Full Adders [5]. First, the Schematic of the RCA was made, and then the transistor-level layout of the RCA was made, verified through design rule check (DRC) and layout versus schematic (LVS), and simulated in ADE L using RCX extraction.

a) RCA Schematic: Here, the RCA schematic was made using 4 Full Adders (Fig. 3). The inputs are Cin, A0 to A3, and B0 to B3, and the outputs are S0 to S3 and Cout.

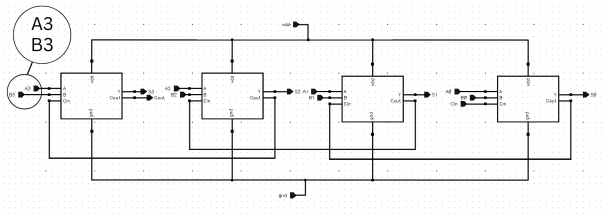


Fig. 3: Schematic of the RCA

b) RCA Layout: Here, the RCA transistor-level layout was made using standard cell NAND gate cells (Fig. 4). The DRC and LVS check was done, and finally, it was ready for RCX extraction and simulation.

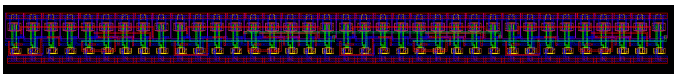


Fig. 4: Layout of the RCA

B. 4-Bit Carry-Look-Ahead Adder (CLA):

The CLA was made with AND, OR, and XOR gates for parallel carry computation [13], [14]. First, AND gates were made with NAND gates and inverters, and XOR Gates were made with NAND Gates only. Then the Carry Generation Circuit was made with these AND, XOR, and OR Gates. After that, the Full CLA was made with the Carry Generation Circuit. It calculates Carry Generate (G_i) and Carry Propagate (P_i) signals for each bit position.

- $G_i = A_i \cdot B_i$
- $P_i = A_i \oplus B_i$

The final carry equations for each bit position are then:

- $C_1 = G_0 + P_0 \cdot C_0$
- $C_2 = G_1 + P_1 \cdot G_0 + P_1 \cdot P_0 \cdot C_0$
- $C_3 = G_2 + P_2 \cdot G_1 + P_2 \cdot P_1 \cdot G_0 + P_2 \cdot P_1 \cdot P_0 \cdot C_0$
- $C_4 = G_3 + P_3 \cdot G_2 + P_3 \cdot P_2 \cdot G_1 + P_3 \cdot P_2 \cdot P_1 \cdot G_0 + P_3 \cdot P_2 \cdot P_1 \cdot P_0 \cdot C_0$

Authorized licensed use limited to: INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY. Downloaded on October 25,2026 at 13:34:52 UTC from IEEE Xplore. Restrictions apply.

1) XOR Gate Design and Implementation: The XOR Gate was made using the 4 NAND Gates (Fig. 5). The inputs are A and B, and the output is Y.

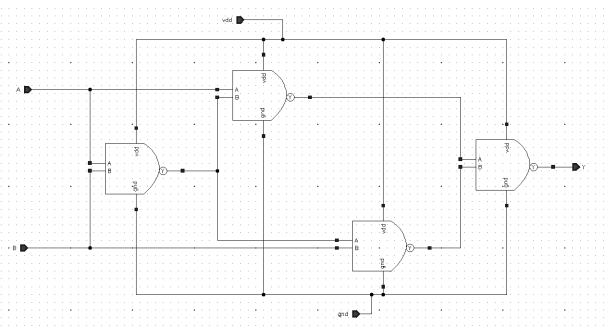


Fig. 5: Schematic of the XOR Gate

2) Carry Generation Circuit Design and Implementation: First, the Schematic and then the transistor-level layout were made. The Schematic of the Carry Generation Circuit was made by AND gates and OR gates (Fig. 6).

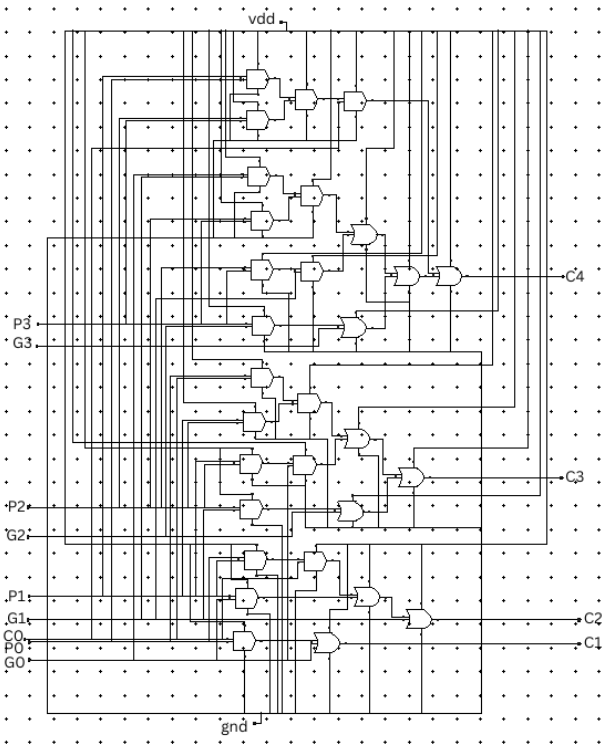


Fig. 6: Schematic of the Carry Generation Circuit

3) CLA Design and Implementation: First, the Schematic and then the transistor-level layout were made, verified through design rule check (DRC) and layout versus schematic (LVS), and simulated in ADE L using RCX extraction.

a) CLA Schematic: The Schematic was made with the Carry Generation Circuit, AND gates, and XOR gates (Fig. 7). It is very complicated compared with the RCA. There are 9 inputs in the CLA from A0 to A3 and B0 to B3, and Cin. And there are 5 outputs, S0 to S3, and then Cout. Even though it is a really complicated circuit, it is much faster than RCA.

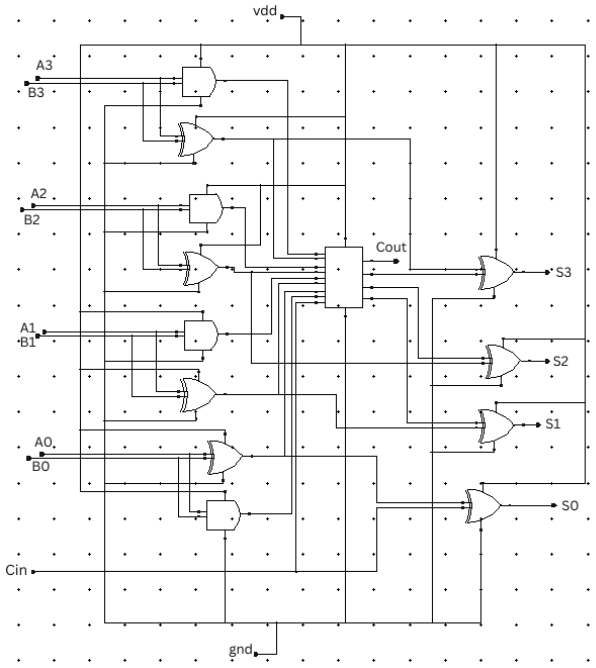


Fig. 7: Schematic of the CLA

b) CLA Layout: Here, the CLA transistor-level layout was made using standard cells (Fig. 8). The DRC and LVS check was done, and finally, it was ready for RCX extraction and simulation.

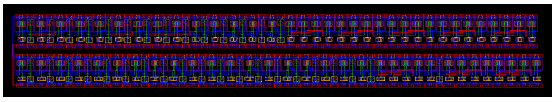


Fig. 8: Layout of the CLA

c) CLA Simulation: Here, the CLA simulation is done using ADE L (Fig. 9). The simulation was done for all the possible 512 combinations. From A0 to A3 and B0 to B3, and Cin, a total of 9 inputs, so there are 2^9 or 512 combinations of inputs.

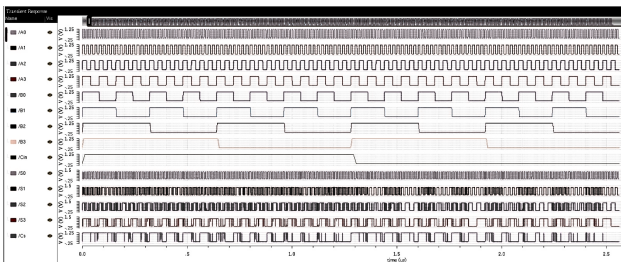


Fig. 9: Simulation of the CLA

Authorized licensed use limited to: INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY. Downloaded on October 25,2026 at 13:34:52 UTC from IEEE Xplore. Restrictions apply.

III. Simulation Methodology

This section explains the detailed methodology for the simulation and performance calculation of the Ripple-Carry Adder (RCA) and the Carry-Look-Ahead Adder (CLA), both of which were simulated in ADE-L using the GPDK90 CMOS process library.

A. Propagation Delay Measurement:

To accurately calculate the dynamic performance of each adder, both best-case (BC) and worst-case (WC) propagation delays were measured. For all cases:

- V_{DD} = 1.2V
- GND = 0V
- $Threshold$ = 0.6V

1) Simulation Setup: To calculate the delay, the Adders were simulated under different conditions. Simulation setup for different cases is shown in Table I.

TABLE I: Simulation Setup for 4-bit Adders

Adder	Case	Input (4-bit)	Delay Path	Edge
RCA	Worst	A=1111, B=0000, Cin: 0→1	Cin → ripple through 4 FAs → Cout	Cin ↑ (0→1)
RCA	Best	A=0000, B=0000, Cin: 0→1	Cin → S_0 only	Cin ↑ (0→1)
CLA	Worst	A=1111, B=0000, Cin: 0→1	Cin → look-ahead tree → Cout	Cin ↑ (0→1)
CLA	Best	A=1111, B=1111, Cin: 0→1	Cin → immediate Generate → S_0	Cin ↑ (0→1)

2) Delay Equations: After the simulation, delay equations were used to calculate the delays in the Calculator in ADE L. Different equations are shown in Table II.

TABLE II: Delay Equations for 4-bit Adders

Adder	Case	Delay Equation	Notes / Setup Details
RCA	Best	delay(?wf1 VT("/Cin"), ?value1 0.6, ?edge1 "falling", ?nth1 2, ?td1 0.0, ?wf2 VT("/S0"), ?value2 0.6, ?edge2 "falling", ?nth2 2)	Input: A=0000, B=0000, Cin=0→1; measure Cin → S0
RCA	Worst	delay(?wf1 VT("/Cin"), ?value1 0.6, ?edge1 "rising", ?nth1 2, ?td1 0.0, ?wf2 VT("/Cout"), ?value2 0.6, ?edge2 "rising", ?nth2 2)	Input: A=1111, B=0001, Cin=0→1; measure ripple through 4 adders
CLA	Best	delay(?wf1 VT("/Cin"), ?value1 0.6, ?edge1 "rising", ?nth1 2, ?td1 0.0, ?wf2 VT("/S0"), ?value2 0.6, ?edge2 "rising", ?nth2 2)	Input: A=1111, B=1111, Cin=0→1; measure Cin → S0
CLA	Worst	delay(?wf1 VT("/Cin"), ?value1 0.6, ?edge1 "rising", ?nth1 2, ?td1 0.0, ?wf2 VT("/Cout"), ?value2 0.6, ?edge2 "rising", ?nth2 2)	Input: A=1111, B=0000, Cin=0→1; measure through CLA tree

B. Power Measurement:

The main equation that is used to measure the Average Power for both adders, RCA and CLA, is [**?**]:

Average Power = average (getData (":pwr" ? result " tran "))

C. Area Measurement and Transistor Count:

The total silicon area and transistor count were directly measured from the verified transistor-level layouts of both Adders.

D. Power-Delay Product (PDP):

PDP basically means the energy that is consumed per operation. It is calculated with the equation below:

$$PDP = \text{Average Power} \times \text{delay time}$$

E. Area-Delay Product (ADP):

ADP is calculated with the equation below:

$$ADP = \text{Area} \times \text{delay time}$$

IV. Results and Analysis

This section presents the final results and analysis of the simulation obtained by the 4-bit Ripple-Carry Adder (RCA) and the Carry-Look-Ahead Adder (CLA). The main performance parameters are based on propagation delay, average power consumption, total silicon area, and transistor count. Additionally, more advanced parameters, the Power-Delay Product (PDP) and Area-Delay Product (ADP), were calculated to get a clearer picture of their efficiency.

A. Simulation Results:

In the (Fig. 10), the simulation of RCA is shown for the worst-case delay. Here, the inputs are A=1111, B=0001, Cin= pulse going 0→1. In the (Fig. 11), the simulation of CLA is shown as well for the worst-case delay.

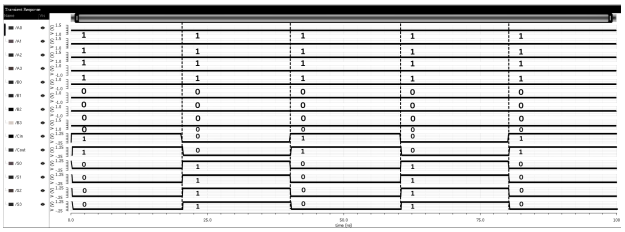


Fig. 10: Simulation of RCA (A=1111, B=0000, Cin=0→1)

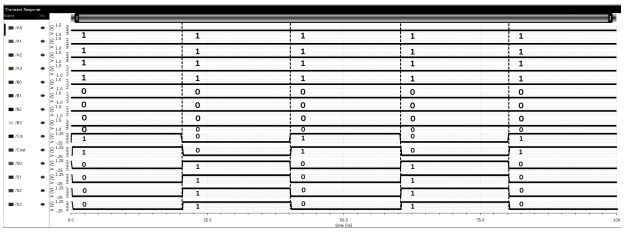


Fig. 11: Simulation of CLA (A=1111, B=0000, Cin=0→1)

Authorized licensed use limited to: INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY. Downloaded on October 25,2026 at 13:34:52 UTC from IEEE Xplore. Restrictions apply.

B. Propagation Delay:

Here are the best and the worst-case propagation delays for RCA and CLA shown in Table III and Fig. 12.

TABLE III: Best and Worst Case Delays for 4-bit Adders

Adder	Best Case Delay (ps)	Worst Case Delay (ps)
RCA	50.38	255.9
CLA	50.38	79.7

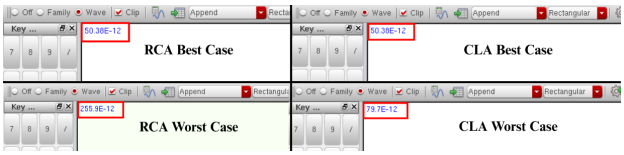


Fig. 12: Best and Worst-case delays of RCA and CLA

C. Average Power Consumption:

The data below in Table IV and Fig. 13 shows the Average power consumption of the two adders, RCA and CLA.

TABLE IV: Average Power Consumption of 4-bit Adders

Adder	Average Power (µJ)
RCA	4.044
CLA	8.326



Fig. 13: Average Power Consumption of RCA and CLA

D. Area and Transistor Count:

The total silicon area and transistor count were extracted directly from the verified layouts of both Adders, shown in Table V.

TABLE V: Silicon Area and Transistor Count for 4-bit Adders

Adder	Area (μm^2)	Transistors
RCA	337	144
CLA	797	332

E. Power-Delay Product (PDP) and Area-Delay Product (ADP):

The data below in Table VI shows the Power-Delay Product (PDP) and Area-Delay Product (ADP) of the two adders, RCA and CLA.

TABLE VI: Worst-Case PDP and ADP for 4-bit Adders

Adder	Worst-Case PDP (J·s)	Worst-Case ADP (μm^2·s)
RCA	1.03×10^{-15}	8.62×10^{-8}
CLA	6.64×10^{-16}	6.35×10^{-8}

F. Comparison Table and Graph:

The data below in Table VII and Fig. 14 shows the overall comparison of RCA and CLA. All the data are combined here for further analysis.

TABLE VII: Comparison of 4-bit RCA and CLA

Metric	RCA	CLA
Best Case Delay (ps)	50.38	50.38
Worst Case Delay (ps)	255.9	79.7
Average Power (µJ)	4.044	8.326
Area (μm^2)	337	797
Transistors	144	332
Power-Delay Product (PDP, J·s)	1.03×10^{-15}	6.64×10^{-16}
Area-Delay Product (ADP, μm^2·s)	8.62×10^{-8}	6.35×10^{-8}

In Fig. 14, the graphs of the overall comparison of RCA and CLA are shown. Black here represents RCA, and red represents CLA.

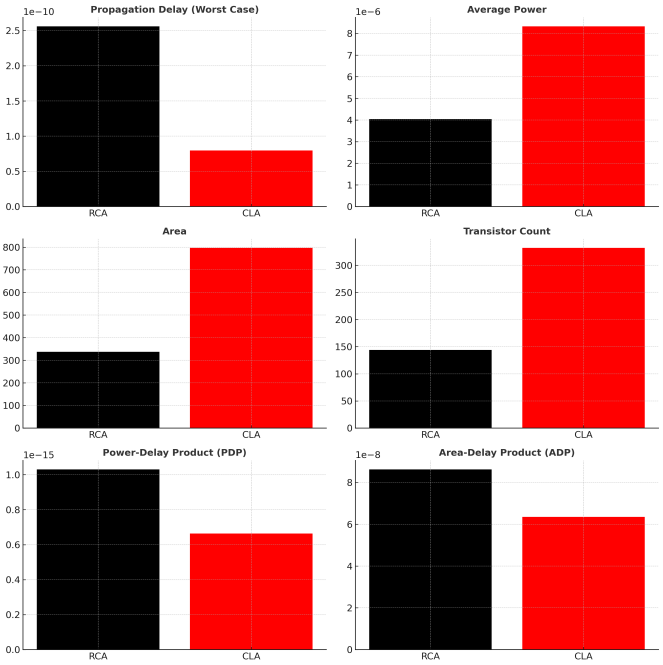


Fig. 14: Comparing all the results for RCA and CLA

V. Discussion on Results

The primary objective of this study was to show a clear comparison of the design trade-offs between the Ripple Carry Adder (RCA) and the Carry Look-Ahead Adder (CLA) in the GPDK90 CMOS process and to find out the fundamental differences in their architectures, serial vs. parallel carry propagation.

As expected, the CLA is approximately 3.2 times faster than the RCA, with a worst-case delay of 79.7 ps compared to the RCA's 255.9 ps. However, this performance gain of CLA comes with a cost. The RCA has a significantly smaller silicon area of 337 μm^2 and a lower transistor count of 144. But the CLA requires a larger area of 797 μm^2 and more

Authorized licensed use limited to: INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY. Downloaded on October 25,2026 at 13:34:52 UTC from IEEE Xplore. Restrictions apply.

than double the number of transistors of 332, to implement its complex logic, which increases the physical complexity and also leads to a higher average power consumption for the CLA (8.326×10^{-6} J) compared to the RCA (4.044×10^{-6} J). A good design must balance multiple objectives. A good design must be efficient, which means even if the design is consuming more power or taking more space, it has to utilize that extra power or space efficiently. The PDP and ADP show us a balanced evaluation between the two adders. The CLA shows us the PDP of 6.64×10^{-16} J·s, which is way better than the RCA's PDP of 1.03×10^{-15} J·s. This shows that despite consuming more power, CLA can easily perform a single computation with greater energy efficiency. Similarly, the CLA shows a lower ADP, which is 6.35×10^{-8} $\mu m^2 \cdot s$, and it shows that the CLA provides more performance per unit of silicon area.

VI. Conclusion and Future Work

This paper presented the schematic and transistor-level design of a 4-bit Ripple Carry Adder (RCA) and a 4-bit Carry Look-Ahead Adder (CLA), as well as a comparative performance analysis between them, implemented using the GPDK90 CMOS process. The study successfully quantified the fundamental design and implementation trade-offs between these two widely used adder architectures, RCA and CLA. The simulations and results show that the CLA is a way faster Adder architecture for any high-performance applications. While this speed, the lower (3.2 times) worst-case delay of the CLA always comes at the expense of a larger silicon area, which is almost 2 times, significantly higher transistor count, which is again almost 2 times, and finally, greater, yet again almost double average power consumption. The CLA's lower PDP means it performs a single task with greater energy efficiency. The lower PDP and ADP compared with RCA demonstrate that its speed benefits outweigh these costs, making it a more suitable choice for designs where performance is the main objective. On the other hand, the RCA still remains a simple, compact, and power-efficient alternative, making it ideal for low-power or area-constrained designs where the added latency or the delay of the adder is not a big problem. This study validates that the choice between these two architectures must be a design decision based on the specific application's requirements.

Future work can be done on this analysis for larger bit-widths like 8-bit, 16-bit, etc, or explore different adder architectures such as the Carry-Select or Carry-Skip adders, or Parallel Prefix Adders like the Kogge-Stone or Brent-Kung designs, or investigate the impact of alternative logic styles on these architectures and calculate different trade-offs. For scalability, 16, 32, or 64-bit adders can be made. But it has to be done by HDL coding and RTL via software like openROAD, where the layout is generated automatically from HDL code, cause it is too complex to do it manually by hand. Future work can be done to optimize this into a full custom design for these larger bit widths to extract the maximum performance out of these designs.

References

[1] Jan M Rabaey, Anantha P Chandrakasan, and Borivoje Nikolic. The devices. *Digital integrated circuits: a design perspective, 2nd edn. Pearson Education, New York*, 2003.

[2] Neil HE Weste and Kamran Eshraghian. *Principles of CMOS VLSI design: a systems perspective*. Addison-Wesley Longman Publishing Co., Inc., 1985.

[3] Sarah Harris and David Harris. *Digital Design and Computer Architecture, RISC-V Edition*. Morgan Kaufmann, 2021.

[4] Pak K Chan and Martine DF Schlag. Analysis and design of cmos manchester adders with variable carry-skip. *IEEE Transactions on Computers*, 39(8):983–992, 2002.

[5] J. A. Emmanuel and A. Ahmed. Seminal paper ripple carry adder. *International Journal of Engineering Technology Research*, 2018.

[6] G Karthik Reddy and D Sharat Babu Rao. A comparative study on low-power and high speed carry select adder. In *2015 IEEE 9th International Conference on Intelligent Systems and Control (ISCO)*, pages 1–7. IEEE, 2015.

[7] Massoud Pedram. Power optimization and management in vlsi design: A survey. *ACM Transactions on Design Automation of Electronic Systems*, 1(1):3–56, 1996.

[8] MUDRABOYINA SRINIVASA RAO and BELLAM VARALAKSHMI. Performance analysis of high speed cmos full adder circuits for embedded system. *International Journal of Computer Engineering In Research Trends*, 2(11):857–861, 2015.

[9] Sarvesh Ganesan. *Area, delay and power comparison of adder topologies*. PhD thesis, 2015.

[10] R. A. Emmanuel and A. A. Ali. Comparative analysis of adders vlsi. *International Journal of Engineering Technology Research*, 2016.

[11] Kumar Mahesh and S. J. Murthy. Area-delay and power comparison of adder topologies. *ResearchGate*, 2015.

[12] S. Gaur. Comparative analysis of adders. *International Journal of Science Research (IJSR)*, 4(3):2205–2207, 2015.

[13] Anthony N. Emmanuel and D. A. Ahmed. Rca and cla performance comparison paper. *arXiv preprint arXiv:1810.01115*, 2018.

[14] R. A. Emmanuel and A. Ahmed. Seminal paper carry-select adder. *ER Publication*, 2016.

Authorized licensed use limited to: INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY. Downloaded on October 25,2026 at 13:34:52 UTC from IEEE Xplore. Restrictions apply.